Software Process Management

Lecture 10

8th Step Process – Elicitation

- 5th step – Elicitation
- The elicitation step is intended to collect all information necessary for actually describing the target software process.
- Mainly, the necessary information comprises:
- -The process entities. These include activities, roles, work products, tools, and, potentially, many others. Oftentimes, entities form hierarchies.
- For example, major activities can be decomposed into smaller tasks, and tasks, in turn, can be decomposed into individual steps.

Elicitation

- - Relationships between entities.
- For example, information about which activities produce or consume which products, which tools are necessary for performing a task, or which tasks compose an activity is expressed through relationships.

- – Behavioral properties of the process entities,
- e.g., which conditions must hold in order for an activity to be started (preconditions), or which criteria a work product must fulfill in order to be accepted.

- The techniques used to elicit information are a central success factor for a descriptive process modelling effort.
- Depending on the organization's culture, for example, either individual interviews, group workshops, or a combination of both may be the best option for obtaining direct information from process participants.

- Importantly, never judge the present methods or people who perform these methods identified during the descriptive process model development.

6) Create the Process Model

- The information collected during the elicitation step can be used to create a model using the chosen modelling notation.
- It is advisable to collect the information in the following general order.



- - Start by modelling products, because they are usually the easiest process entities to identify. In contrast to activities, which are often only implicitly defined, a set of products usually explicitly exists in an organization and can be directly modelled.
- - Continue with the activities and the product flow.

- - When activities are modeled, attach associated roles to the activities or the products.
- – Model further entity types, such as resources, as needed.
- – As a last step, model behavioral restrictions associated with entities, such as the preconditions that must be met before an activity can start.

- Once an initial model is available based on elicited information, it must be reviewed by the process participants.
- Such a review will often result in additional elicitation activities and successive model refinements.

7) Analyse the Process Model

- As process models become complex, there is an increasing risk of defects being inadvertently introduced into the model by the process engineers.
- The purpose of the process model analysis step is to detect such defects and correct them.
- Depending on the notation used, there are many possible types of defects that may arise in practice.
- The following list contains a few examples of possible process model defects:

- – Dangling reference: An entity references another, undefined entity, e.g., an activity description mentions an input work product that has no definition of its own in the model.



- – Inconsistent precondition: The start conditions stated for an activity are contradictory and cannot be fulfilled in practice.

- – Inconsistent references: The same entity is mentioned in the model using different, inconsistent names.

- – Orphan entity: An entity is defined in the model, but is not referenced by any other entity in a meaningful way.

- – Dependency cycle: An activity depends on a particular input product in order to start, but, according to the model, this input product cannot be produced unless the activity has already been finished.

- – Incomplete descriptions: Important entity attributes, such as activity or product descriptions, are missing or empty in some or all of the process entities.

Types of Analysis

- – Completeness analyses.
- Even if a model correctly describes a process, the level of detail or the amount of information in the description could be insufficient.
- Completeness analyses are concerned with making sure that all relevant information is available and at a sufficient level of detail.
- Checking for empty fields in entity description forms, and making sure that all activities in a process are refined to the level of individual tasks, are two examples of possible completeness analyses.

- – Consistency analyses.
- This type of analysis is related to making sure that elements of the model do not contradict each other.
- An example of a consistency analysis is to make sure that all products mentioned by the natural language description of an activity are modelled explicitly as inputs or outputs of the activity, and that they are mentioned using the same name used in the corresponding product description.

- – Dynamic analyses.
- These analyses are concerned with problems that may arise when the process described by the model is executed.
- A typical dynamic analysis is to search for potential deadlocks that may occur while executing the process.

- Depending on the levels of formality and detail of the chosen process model notation, process analysis can be automated.
- In many cases however, automation is not possible due to the lack of formal models used in.

8) Analyse the process

- Final step – Using the developed process model to analyse the process.
- i.e., to use the process model to track or analyze process performance, depending on the process modeling objectives stated in step 1.
- One possibility is to track the process by asking process performers to log, for example, the start and finish times for their activities.
- Another option is to make “snapshots” by asking people about their current activities at regular intervals.
- The resulting data can be used for qualitative or for quantitative analyses.

- Qualitative analyses aim at identifying weaknesses of the processes, e.g., when too many responsibilities are pinned to a single role, or when too many roles are assigned to a single person, or when feedback loops become so excessive that they consume more time than productive project work.

- Quantitative analyses aim at identifying correlations between process attributes, e.g., when the number of requirements is more than 30% higher than average, the number of defects rises by 70%.

DPM Alternatives

- There are alternative modelling approaches than the 8 step process that we discussed.

- 1) Multi View Process Modelling



- 2) Elicit

Multi View Process Modelling

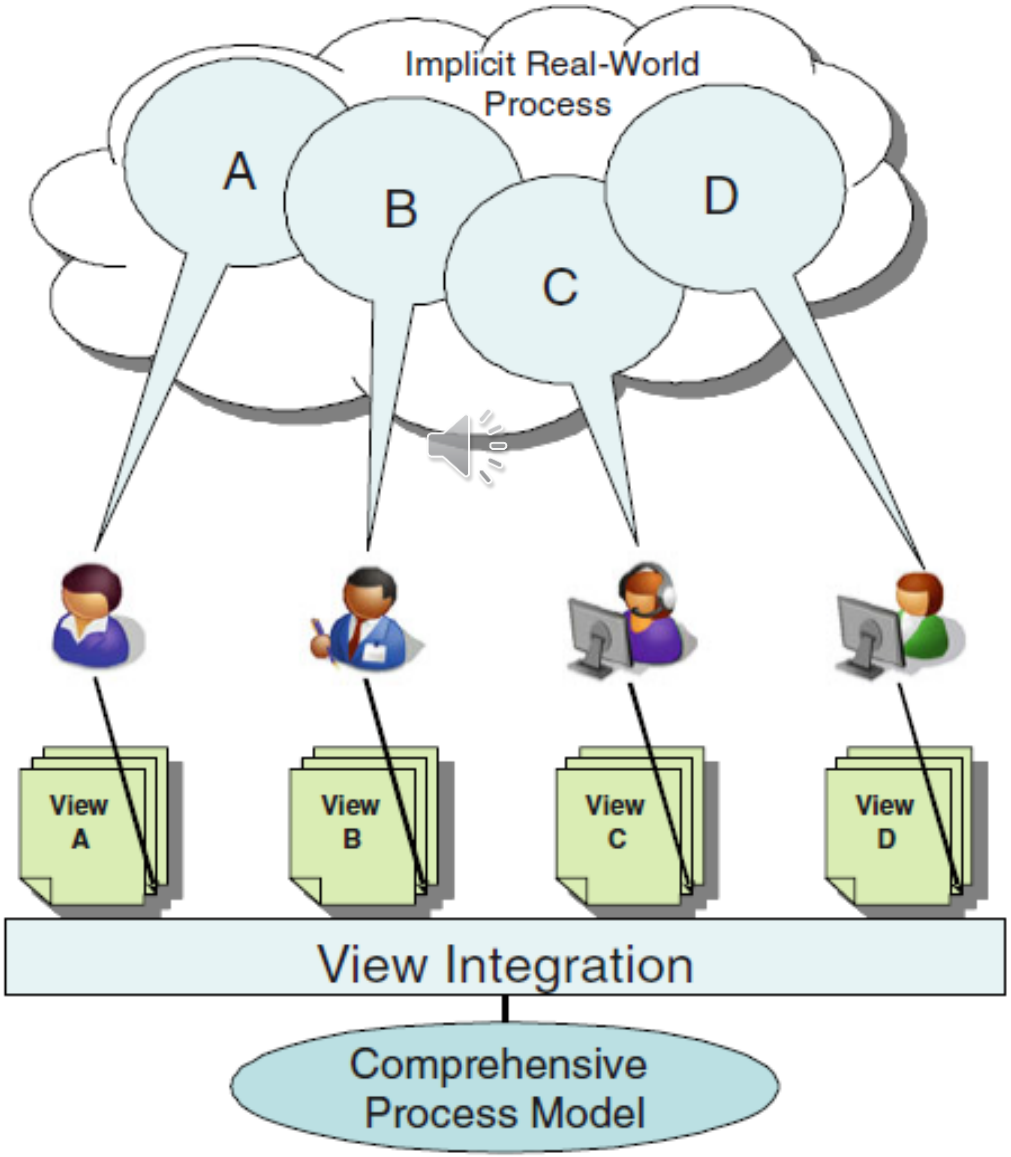
- This method replaces steps 5 and 6 of 8 step process with an approach that collects and integrates multiple views on the same process.
  - provides a specific way to conduct the elicitation step and build a model based on it.



- The main assumption behind multi-view process modelling is that, given the complexity of actual software processes, no single process participant has a complete, unified, and sufficiently detailed view of the process.

- Participants are normally experts in rather narrow areas of the process.

- Multi-view modelling starts by modelling the process as understood by individual participants.
- This results in separate, role-specific process views that contain potentially detailed but partial process information.
- In a later step, the individual views are systematically integrated to form a single, complete process model.
- This procedure is illustrated in Fig.


Implicit Real-World Process
A
B
C
D
View A
View B
View C
View D
View Integration
Comprehensive Process Model

- The integration of role-specific is an iterative process comprising four steps:
- 1. Modelling.
- In the modelling step, role-specific views are created by independently interviewing participants.
- The resulting views are partial process models that contain all activities performed by a role, together with related process entities such as work products or other roles with which the analyzed role interacts.
- Role-specific views are expected to be internally consistent, but they are not guaranteed to be consistent with each other.

- 2. Similarity analysis.
- In this step, the various role-specific views are analyzed to identify common objects, such as shared activities or work products.



- Notice that, due to the fact that each view comes from a different person or group of persons, there may be inconsistencies in naming and other modelling aspects.

- For this reason, a combination of manual and automated techniques is used to find matching entities.

- 3. Inter-view consistency checking.
- Once matching elements have been identified, a number of consistency checks are performed between interrelated views in order to detect potential inconsistencies between them.



- Whenever inconsistencies are found, they are resolved by changing the views as necessary.

- 4. Integration.
- At this point, the separate views are consistent enough to be merged into a single, integrated process model.

- As originally proposed, the Multi-View Modelling method is based on the MVP-L process language.
- In practice, however, most of it is independent of a specific modelling language and can be applied in a quite general way.

- MVP-L – Multi View Process Modelling Language

- Steps 2 and 3 of this method can be very challenging!
- In the case of Similarity Analysis (Step 2) multiple views on entities can be expressed using different terms/expressions.
- At the same time similar terms/expressions can be actually representing different views/entities.

- Since reporting nonmatching entities as matching (false positives) and to missing actual pairs of matching entities (false negatives) all results obtained automatically must be verified manually.

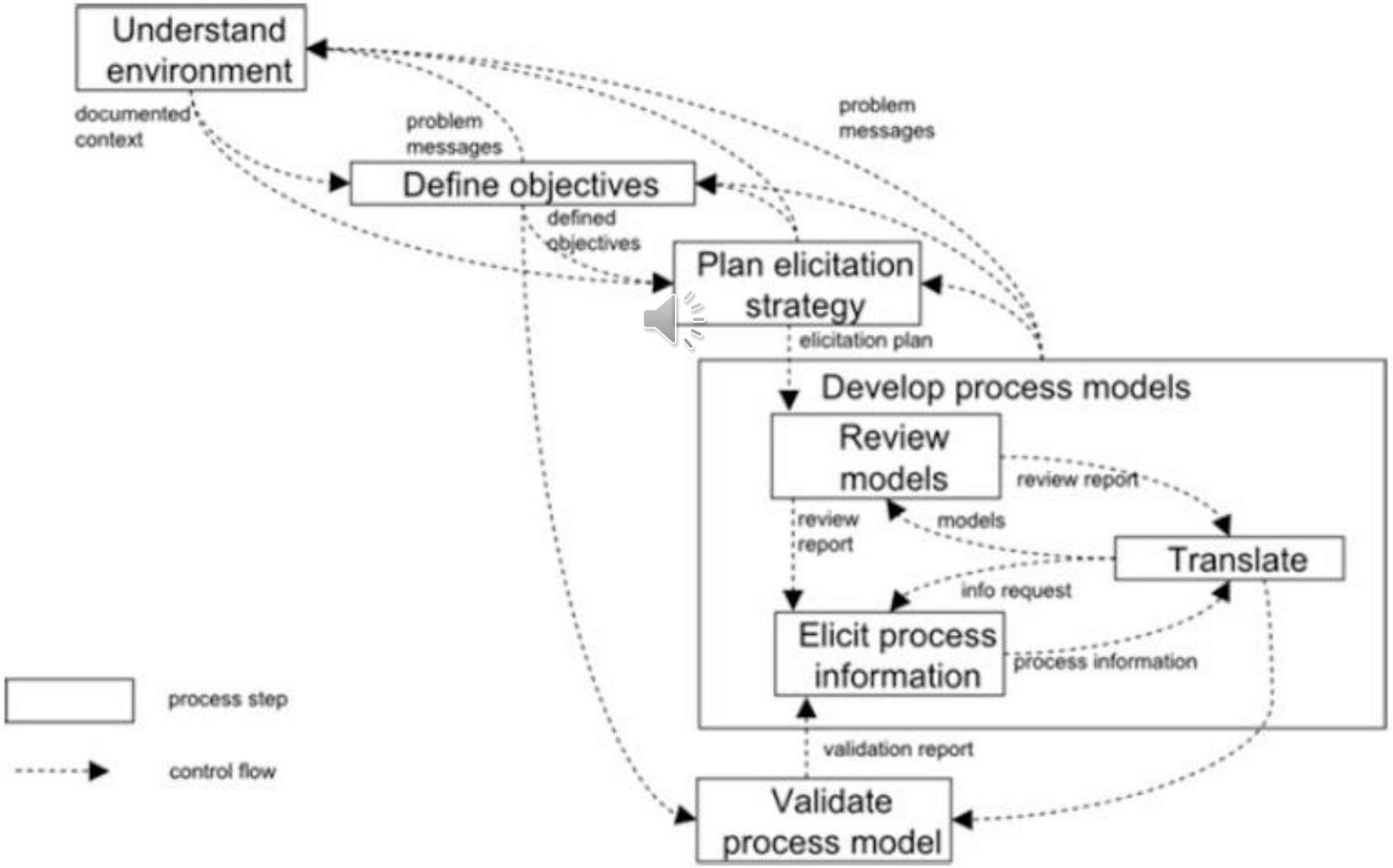
- Regarding inter-view consistency (Step 3), a number of consistency rules can be applied once matching entities have been found.
- For example, matching entities in all views must have the same name, whereas entities that do not match any other entity should have a unique name.
- Depending on the set of modelling concepts used, more advanced rules can be devised.

Elicit

- Elicit is a general framework for process elicitation.
- Its main difference to the general, 8-step approach presented in this chapter is that it does not expect the organization to select a schema, a modelling language, or a set of modelling tools.
- Elicit provides all of these elements, together with a set of methodological steps for using them, thus making its application somewhat simpler.

- Elicit works by looking at the process from a variety of viewpoints, which are later consolidated into a unified view.
- The Elicit schema is structured as a set of five so-called **perspectives**: artefacts, process steps, roles, resources, and constraints;
- and together with three types of **properties**: descriptive, static, and dynamic.
- The combination of a perspective and a property is called a view in Elicit.
- This means that there are 15 possible views: For each of these views, Elicit provides a set of so-called attributes.

Steps of the Elicit Method



1. Understand organizational environment

- This step consists of understanding key aspects of the organization, such as
  - its size,
  - management structure,
  - number and size of the running projects,
  - common roles played by people in projects,
  - etc.

- Understanding these issues is considered fundamental for process engineers to be able to do their job in later steps.

2. Define objectives

- This includes defining the exact scope of the model e.g.,
  - its granularity and the degree of consistency,
  - completeness,
  - accuracy,
  - and clarity that must be achieved
- And setting goals for the modelling effort itself, such as
  - its targeted costs and
  - time frame.

3. Plan the elicitation strategy

- Given the set of goals produced in the previous step, it must be defined how to achieve these goals.
- A key aspect of this step is to allocate appropriate resources for the modelling effort.
- For example, it must be decided
  - which people will be interviewed,
  - when reviews are going to be conducted,
  - etc.

4. Develop process models

- This step involves collecting information according to the Elicit process schema, as explained earlier.
- The collected information is then used to create models using process modelling tools and submitted for review.
- If reviews find deficiencies in the model, further information collection and modelling must be performed.

5. Validate process models

- The main objective of this step is to have the models validated by their intended recipients.
- This includes checking that the models represent the actual process in an appropriate way, and, generally, validating the models against the modelling objectives stated originally.

- Elicit Method
- Use feedback steps as learning steps from other steps
- Suggests tool usage for some cases

Managing Descriptive Process Modelling Risks

- Managing Resistance of Participants
- Managing Inaccurate Reporting Challenges
- Underestimating the required investments
- Underestimating the complexity of process model